本资料为2023年复变作业册积分变换解析，由于流传下来的复变作业册解析无积分变换部分，而作者正好在学习本部分，故想要亲自填补这一空缺，以为后来人提供一份参考。在学导团和治学团志愿者们的热心帮助下，我们完成了编写，由于编写时间仓促，难免出现错误或者遗漏，还请谅解，同时也欢迎大家与我们取得联系，修正错误 。

在此感谢以下志愿者对本资料的贡献:
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Fourier

7.1

$$\mathscr{F}[f(t)]=\int_{-\infty}^{+\infty} f(t)e^{-j\omega t}\mathrm{d}t=\int_{-1}^{1} e^{-j\omega t}\mathrm{d}t$$

因为:

$$e^{-j\omega t}=\cos(j\omega)-\sin(j\omega)$$

上式

$$=\frac{2\sin(\omega)}{\omega}$$

选A

7.2

$$F(\omega)=\int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

所以

$$\int_{-\infty}^{\infty} f(t)\cos\omega_0 t e^{-i\omega t}dt = \frac{1}{2}[\int_{-\infty}^{\infty} f(t)te^{-i(\omega+\omega_0)t}dt + \int_{-\infty}^{\infty} f(t)te^{-i(\omega-\omega_0)t}dt]$$
$$= \frac{1}{2}[F(\omega+\omega_0) + F(\omega-\omega_0)]$$

故选 A

7.3

$$\mathscr{F}[\sin 2t] = \mathscr{F}[\frac{e^{2it}-e^{-2it}}{2i}] = \frac{2\pi[\delta(\omega-2)-\delta(\omega+2)]}{2i} = \pi i[\delta(\omega+2)-\delta(\omega-2)]$$

7.4

$$\int_{-\infty}^{\infty} f(t)e^{\omega_0 t}e^{-i\omega t}dt = \int_{-\infty}^{\infty} f(t)e^{(\omega_0-i\omega)t}dt$$

因为

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$
$$F(\omega+\omega_0) = \int_{-\infty}^{\infty} f(t)e^{-i(\omega+\omega_0)t}dt$$

显然不等，故选C

7.5

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{+\infty} f(t)\, e^{-j\omega t}\mathrm{d}t \\ &= \int_{0}^{+\infty} \sin 2t\, e^{-(j\omega+1)t}\mathrm{d}t \\ &= \frac{2}{(j\omega+1)^2+4} \\ &= \frac{2}{-\omega^2+2\omega j+5} \end{aligned}$$

则:

$$\begin{aligned} f(t) &= \frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{2}{-\omega^2+2\omega j+5}\, e^{-j\omega t}\mathrm{d}t \\ &= \frac{1}{\pi}\int_{-\infty}^{+\infty} \frac{e^{-j\omega t}}{-\omega^2+2\omega j+5}\mathrm{d}t \end{aligned}$$

7.6

$$\mathscr{F}[f(t)] = \int_{-\infty}^{\infty} e^{-|t|} e^{-jwt}\, dt = \int_{-\infty}^{0} e^{(1-jw)t}\, dt + \int_{0}^{\infty} e^{-(1+jw)t}\, dt$$

$$= \frac{2}{1+w^2}$$

则

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2}{1+\omega^2} e^{j\omega t} d\omega = \frac{2}{\pi} \int_{0}^{\infty} \frac{\cos \omega t}{1+\omega^2} d\omega$$

则

$$\int_{0}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{2} f(1) = \frac{\pi}{2e}$$

7.7

$$\begin{aligned}
\mathscr{F}[f(t)] &= \int_{0}^{+\infty} e^{-\beta t} \cdot e^{-j\omega t} \mathrm{d}t \\
&= \int_{0}^{+\infty} e^{-j\omega t - \beta t} \mathrm{d}t \\
&= \frac{1}{\beta + j\omega} \\
&= \frac{\beta - j\omega}{\beta^2 + \omega^2}
\end{aligned}$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\beta - j\omega}{\beta^2 + \omega^2} e^{-j\omega t} \mathrm{d}\omega = \begin{cases} 0, & t < 0 \\ \frac{1}{2}, & t = 0 \\ e^{-\beta t} & t > 0 \end{cases}$$

令$\beta = 1, t = 0$ 得:

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{1 - j\omega}{1^2 + \omega^2} \mathrm{d}\omega = \frac{1}{\pi} \int_{0}^{+\infty} \frac{1}{\omega^2 + 1} \mathrm{d}\omega$$

则:

$$\int_{0}^{+\infty} \frac{1}{x^2 + 1} \mathrm{d}x = \frac{\pi}{2}$$

7.8

$$
\begin{aligned}
f(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\pi[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)]e^{j\omega t}d\omega \\
&= \frac{1}{2}[\int_{-\infty}^{\infty}\delta(\omega+\omega_0)e^{j\omega t}d\omega+\int_{-\infty}^{\infty}\delta(\omega-\omega_0)e^{j\omega t}d\omega] \\
&= \frac{1}{2}[e^{-j\omega_0 t}+e^{j\omega_0 t}] = \cos(\omega_0 t)
\end{aligned}
$$

7.9

$$
\begin{aligned}
f_1(t)*f_2(t) &= \int_0^t \sin\tau e^{-(t-\tau)}\mathrm{d}\tau \\
&= e^{-t}\int_0^t \sin\tau e^{\tau}\mathrm{d}\tau \\
&= e^{-t}\frac{\sin\tau-\cos\tau}{2}e^{\tau}|_0^t \\
&= e^{-t}[\frac{\sin t-\cos t}{2}e^t+\frac{1}{2}] \\
&= \frac{\sin t-\cos t+e^t}{2}
\end{aligned}
$$

Laplace

8.1

$$
\mathscr{L}[\sin t] = \frac{1}{s^2+1}
$$

$$
则\mathscr{L}[\sin(t-\frac{\pi}{3})] = \frac{1}{s^2+1}e^{-\frac{\pi s}{3}}
$$

8.2

对$\mathscr{L}[\sin t]$加入微分环节，得到

$$
\mathscr{L}[f(t)] = \frac{2s}{(s^2+1)^2}
$$

8.3

同时加入微分和延迟环节，容易得到结果为 D

8.4

直接进行Laplace逆变换容易得到答案为 A

8.5

(1)

$$\mathscr{L}[f(t)] = \frac{2(s+1)}{[(s+1)^2+1]^2}$$

(2)

$$\mathscr{L}[\frac{\sin 2t}{t}] = \int_s^{\infty} \frac{2}{s^2+4} ds = \frac{\pi}{2} - arctan\frac{s}{2}$$

$$\mathscr{L}[e^{-t}\frac{\sin 2t}{t}] = \frac{\pi}{2} - arctan\frac{s+1}{2}$$

$$\mathscr{L}[\int_0^t e^{-t}\frac{\sin 2t}{t}] = \frac{1}{s}(\frac{\pi}{2} - arctan\frac{s+1}{2})$$

8.5

(3)

$$\mathscr{L}[f(t)] = \int_0^2 3e^{-st}dt + \int_2^4 -e^{-st}dt = \frac{3+e^{-4s}-4e^{-2s}}{s}$$

(4)

由于$f(t+2\pi) = f(t)$

$$\begin{aligned}
\mathscr{L}[f(t)] &= \sum_{k=0}^{\infty}\int_{2k\pi}^{2(k+1)\pi} f(t)e^{-st}dt \\
&= \sum_{k=0}^{\infty}\int_0^T f(\tau+kT)e^{-s(\tau+kT)}d\tau \\
&= \sum_{k=0}^{\infty} e^{-skT}\int_0^T f(\tau+kT)e^{-s\tau}d\tau \\
&= \frac{1}{1-e^{-sT}}\int_0^{\tau} f(t)e^{-st}dt \\
&= \frac{1}{1-e^{-2\pi s}}\int_0^{\pi} e^{-st}\sin t dt \\
&= \frac{1+e^{-s\pi}}{(1-e^{-2s\pi})(1+s^2)}
\end{aligned}$$

8.6

证明如下：

由于$F(s)=\int_0^{\infty}f(t)e^{-st}dt$ 易得$F^{(n)}(s)=(-1)^n\mathscr{L}[t^nf(t)],Re(s)>c$

(1)

$$\mathscr{L}[e^{-3t}\sin 2t]=\frac{2}{(s+3)^2+4}$$

$$\mathscr{L}[\int_0^t e^{-3t}\sin 2t]=\frac{1}{s}\frac{2}{(s+3)^2+4}$$

$$\mathscr{L}[t\int_0^t e^{-3t}\sin 2t]=\frac{2(3s^2+12s+13)}{s^2[(s+3)^2+4]^2}$$

(2)

因为$F^{(1)}(s)=-\mathscr{L}[tf(t)]$,则$\mathscr{L}^{-1}[F^{(1)}(s)]=-t\mathscr{L}^{-1}[F(s)]$

$$则,f(t)=-\frac{1}{t}\mathscr{L}^{-1}[\frac{1}{s+1}-\frac{1}{s-1}]=\frac{1}{t}(e^t-e^{-t})$$

8.7

(1)

$$F(s)=\frac{2}{5}\frac{1}{s+2}+\frac{3}{5}\frac{1}{s-2}$$

故

$$\mathscr{L}^{-1}[F(s)]=\frac{2}{5}\mathrm{e}^{-3t}+\frac{3}{5}\mathrm{e}^{2t}$$

(2)

$$\begin{aligned}\mathscr{L}^{-1}[F(s)]&=\mathscr{L}^{-1}[\frac{1}{s^2+1}\cdot\frac{1}{s^2+1}]\\&=\frac{1}{4}(\sin 2t*\sin 2t)\\&=\frac{1}{16}(\sin 2t-2t\cos 2t)\end{aligned}$$

(3)

因为

$$\begin{aligned}F(s)=\frac{1}{(s^2+a^2)(s^2-a^2)}&=\frac{1}{2a^2}(\frac{1}{s^2-a^2}-\frac{1}{s^2+a^2})\\&=\frac{1}{2a^2}[\frac{1}{2a}(\frac{1}{s-a}-\frac{1}{s+a})-\frac{1}{s^2+a^2}]\end{aligned}$$

所以

$$\mathscr{L}^{-1}[F(s)]=\frac{1}{4a^3}(\mathrm{e}^{at}-\mathrm{e}^{-at})-\frac{1}{2a^2}\frac{1}{s^2+a^2}$$

(4)

因为

$$F(s)=\frac{s+2}{(s+2)^2+1}\cdot\frac{1}{(s+2)^2+1}$$

所以

$$\mathscr{L}^{-1}[F(s)]=\mathrm{e}^{-2t}\cos t*\mathrm{e}^{-2t}\sin t=\mathrm{e}^{-2t}\frac{t\sin t}{2}$$

(5)

$$\begin{aligned}\mathscr{L}^{-1}[F(s)]&=\mathscr{L}^{-1}[\frac{2s^2+s+5}{(s+1)(s+2)(s+3)}]\\&=\mathscr{L}^{-1}[\frac{3}{s+1}+\frac{-11}{s+2}+\frac{8}{s+3}]\\&=3\mathrm{e}^{-t}-11\mathrm{e}^{-2t}+8\mathrm{e}^{-3t}\end{aligned}$$

8.8

(1)

$$\sin kt*\sin kt=\frac{1}{2}(\frac{\sin kt}{k}-t\cos kt)$$

(2)

$$u(t-a)*f(t)=\int_0^t u(t-a)f(t-\tau)\mathrm{d}\tau\begin{cases}\int_0^t f(\tau)d\tau,a\leq 0\\ \int_a^t f(t-\tau)d\tau,0\leq a\leq t\\ 0,0<t<a\end{cases}$$

8.9

(1)

$$\mathscr{L}[1-\cos t]=\frac{1}{s}-\frac{1}{s^2+1}$$

$$F(s)=\mathscr{L}[\frac{1-\cos t}{t}]=\int_s^{\infty}\frac{1}{s}-\frac{s}{s^2+1}\mathrm{d}s=-\ln\frac{s}{\sqrt{s^2+1}}$$

所以

$$\int_0^{\infty}\frac{1-\cos t}{t}\mathrm{e}^{-t}\mathrm{d}t=F(1)=\frac{1}{2}\ln 2$$

(2)

$$\mathscr{L}[\mathrm{e}^{-t}-\mathrm{e}^{-2t}]=\frac{1}{s+1}-\frac{1}{s+2}$$

$$F(s)=\mathscr{L}[\frac{\mathrm{e}^{-t}-\mathrm{e}^{-2t}}{t}]=\int_s^{\infty}\frac{1}{s+1}-\frac{1}{s+2}\mathrm{d}s=-\ln\frac{s+1}{s+2}$$

则

$$\int_0^{\infty}\frac{\mathrm{e}^{-t}-\mathrm{e}^{-2t}}{t}\mathrm{d}t=F(0)=\ln 2$$

(3)

$$\mathscr{L}[\sin^2 t]=\mathscr{L}[\frac{1-\cos 2t}{2}]=\frac{1}{2}(\frac{1}{s}-\frac{s}{s^2+4})$$

$$F(s)=\mathscr{L}[\frac{\sin^2 t}{t}]=\frac{1}{2}\int_s^{\infty}\frac{1}{s}-\frac{s}{s^2+4}\mathrm{d}s=-\frac{1}{2}\ln\frac{s}{\sqrt{s^2+4}}$$

$$\int_0^{\infty}\frac{\mathrm{e}^{-t}\sin^2 t}{t}\mathrm{d}t=F(1)=\frac{\ln 5}{4}$$

(4)

$$\mathscr{L}[t\sin 2t]=-(\frac{2}{s^2+4})'=\frac{4s}{(s^2+4)^2}$$

$$\int_0^{\infty}t\mathrm{e}^{-3t}\sin 2t\mathrm{d}t=F(3)=\frac{12}{169}$$

8.10

$$\mathscr{L}^{-1}[F(s)]=\cos at*\frac{1}{a}\sin at=\frac{1}{a}\int_0^t\sin a\tau\cos(at-a\tau)\mathrm{d}\tau=\frac{t}{2a}\sin at$$

8.11

由积分的线性性质易证.

8.12

取Laplace变换得:

$$\begin{cases} sX(s)-1+X(s)-Y(s)=\frac{1}{s-1} \\ sY(s)-1+3X(s)-2Y(s)=\frac{2}{s-1} \end{cases}$$

即

$$\begin{cases} (s+1)X(s)-Y(s)=\frac{s}{s-1} \\ 3X(s)+(s-2)Y(s)=\frac{s+1}{s-1} \end{cases}$$

得

$$\begin{cases} X(s)=\frac{1}{s-1} \\ Y(s)=\frac{1}{s-1} \end{cases}$$

根据初值条件，则有$x(t)=\mathrm{e}^t, y(t)=\mathrm{e}^t$.

8.13

取Laplace变换得:

$$s^2Y(s)+w^2Y(s)=\frac{\mathrm{e}^{-st_0}}{s}$$

即:

$$Y(s)=\frac{e^{-st_0}}{s(s^2+w^2)}$$

由

$$\frac{1}{s(s^2+w^2)}=\frac{1}{w^2}\left(\frac{1}{s}+\frac{-s}{s^2+w^2}\right)$$

得: $\mathscr{L}^{-1}\left(\frac{1}{s}-\frac{s}{s^2+w^2}\right)=u(t)-\cos wt$

则:

$$\begin{aligned} Y(t)&=\mathscr{L}^{-1}\left(\frac{\mathrm{e}^{-st_0}}{s(s^2+w^2)}\right) \\ &=\frac{u(t-t_0)-u(t-t_0)\cos w(t-t_0)}{w^2}. \end{aligned}$$

8.14

记

$$\mathscr{L}\left(f\left(t\right)\right)=F\left(s\right),\ \mathscr{L}\left(h\left(t\right)\right)=H\left(s\right)$$

取Laplace变换得:

$$a\cdot sX\left(s\right)-x\left(0\right)+bX\left(s\right)\mathscr{L}\left(f\left(t\right)\right)=c\mathscr{L}\left(h\left(t\right)\right)$$

则:

$$X\left(s\right)=\frac{cH\left(s\right)+x\left(0\right)}{as+bF\left(s\right)}$$

$$\therefore x\left(t\right)=\mathscr{L}^{-1}\left[\frac{cH\left(s\right)+x\left(0\right)}{as+bF\left(s\right)}\right]$$

8.15

易知$f\left(0\right)=0$.

取Laplace变换得：$F\left(s\right)=\dfrac{a}{s^2}+\dfrac{1}{s^2+1}F\left(s\right)$，即$F\left(s\right)=a\left(\dfrac{1}{s^2}+\dfrac{1}{s^4}\right)$.

则

$$f\left(t\right)=\mathscr{L}^{-1}\left[F\left(s\right)\right]=a\left(t+\frac{1}{6}t^3\right)$$

8.16

将$t=0$代入，得：$-2y'\left(0\right)-2y\left(0\right)=0$，即有$y'\left(0\right)=-2$.

取Laplace变换得:

$-\left(s^2Y\left(s\right)-2s+2\right)'+2\left[\left(sY\left(s\right)-2\right)'-sY\left(s\right)-2\right]+\left(-Y'\left(s\right)-2Y\left(s\right)\right)=0$.

即有：$\left(s+1\right)^2Y'\left(s\right)+4\left(s+1\right)Y\left(s\right)+2=0$

$\therefore Y'\left(s\right)+\frac{4}{s+1}Y\left(s\right)=-\frac{2}{\left(s+1\right)^2}$

则:

$$\begin{aligned}Y\left(s\right)&=\mathrm{e}^{\int-\frac{4}{s+1}\mathrm{d}s}\left(C+\int\frac{-2}{\left(s+1\right)^2}\cdot\mathrm{e}^{\int\frac{4}{s+1}\mathrm{d}s}\right)\\&=\frac{2}{\left(s+1\right)^4}\left[C-\left(\frac{s^3}{3}+s^2+s\right)\right]\end{aligned}$$

$\therefore y\left(t\right)=\mathscr{L}\left[Y\left(s\right)\right]=\frac{1}{9}\mathrm{e}^{-t}\left(t^{3}-6\right)+C$，代入初始条件，得:
$y\left(t\right)=\frac{1}{9}\mathrm{e}^{-t}\left(t^{3}-6\right)+\frac{2}{3}$.